## PART A: Peer Code Audit Checklist Table

| Audit Category | Specific Issue Located in Legacy Code | Line Number(s) | Severity (Low / Med / High / Critical) | Impact / Risk |
|---|---|---|---|---|
| **Global State** | Mutable global variables data_Array, currentCount, maxCapacity | Lines 8–10 | High | Reduces modularity; prevents multiple independent array instances; harder to maintain. |
| **Memory Leak** | Resizing allocates temp but never deletes the old dataArray | Lines 18–22 | Critical | Old heap allocation becomes unreachable, causing a memory leak. |
| **Memory Leak** | No delete[] dataArray before program ternination | End of main() | High | Allocated heap memory is not explicitly released. |
| **Out-of Bounds** | printAll() uses i <= currentCount | printAll() | High | Reads one element beyond the valid data range. |
| **Out-of Bounds** | remove_item _alt() checks only negative indexes | remove_item_ at() | Critical | Indexes such as 99 may cause invalid memory access. |
| **Inefficient Shift** | Nested loop is unnecessary during deletion | remove_item_ at() | Medium | Adds unnecessary loop structure and reduces readability. |

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| **Naming Standards** | Inconsistent naming; add_item, findItem, printAll | Multiple | Low | Violates a consistent naming convention. |
| **Hardcoded Sizing** | Matrix dimensions are fixed at 3x3 | processMatrix() | Medium | Cannot easily process matrices of other dimensions. . |
| **Input Validation** | Operations do not consistently validate array state | Multiple | High | Empty or invalid operations may behave unsafely. |
| **Encapsulation** | Data and operations use global state | Multiple | High | Any function can directly modify shared state. |